

Executive Summary & Recommendations

• Overall Customer Churn

The analysis reveals that approximately **26.5% of customers have churned**, while **73.5% have been retained**. Although the majority of customers remain with the company, losing more than one in four customers represents a significant revenue risk and highlights the need for focused retention initiatives.

• Demographic Insights

Customer churn appears to be largely independent of gender, with male and female customers exhibiting similar churn behavior. However, **Senior Citizens account for a disproportionately high share of churners**, indicating that older customers may face service, pricing, or support challenges that increase their likelihood of leaving. This segment should be considered a high-priority group for targeted retention campaigns.

• Tenure is the Strongest Retention Indicator

Customer tenure shows a clear inverse relationship with churn. Customers in their first few months with the company demonstrate the highest churn rates, while long-tenured customers are significantly more loyal. The charts indicate that churn declines steadily as tenure increases, suggesting that the company's greatest retention challenge lies in the early stages of the customer lifecycle. Strengthening onboarding programs and early customer engagement could substantially reduce churn.

• Contract Type Significantly Influences Churn

Contract duration emerges as one of the strongest predictors of customer retention. Customers on **Month-to-Month contracts contribute the majority of churn cases**, whereas customers on **One-Year and Two-Year contracts exhibit substantially lower churn rates**. The findings suggest that longer contractual commitments create stronger customer loyalty and reduce switching behavior. Incentivizing customers to migrate from monthly plans to longer-term contracts could have a meaningful impact on retention.

• Internet Service Analysis

Internet service type plays a major role in churn behavior. Among all internet service categories, **Fiber Optic customers show the highest churn percentage**, despite representing a large portion of the customer base. In contrast, DSL customers demonstrate comparatively lower churn rates, while customers without internet service show minimal churn. This pattern suggests potential issues related to pricing, service expectations, network quality, or customer satisfaction within the Fiber Optic segment.

• Impact of Security and Support Services

Customers who do not subscribe to value-added services consistently exhibit higher churn rates:

- **Online Security:** Customers without this service show substantially higher churn than those who subscribe.
- **Tech Support:** One of the strongest churn indicators; customers lacking technical support are significantly more likely to leave.
- **Device Protection:** Customers without device protection churn at a much higher rate than those who have the service.
- **Online Backup:** Similar trends indicate improved retention among customers using backup services.

Across these categories, churn rates for customers without these services appear to be multiple times higher than those who subscribe, highlighting the importance of perceived service value and customer support.

• Streaming Services Have Limited Impact

Unlike support-related products, **Streaming TV and Streaming Movies show relatively weak relationships with churn**. Customers with and without these services exhibit similar churn patterns, suggesting that entertainment features alone are not major drivers of customer retention. While these services may enhance customer experience, they do not appear to be primary factors influencing a customer's decision to stay or leave.

• Payment Method Analysis

Payment behavior reveals another important pattern. Customers using **Electronic Check** as their payment method exhibit the highest churn percentage among all payment options. Conversely, customers enrolled in automatic payment methods, such as bank transfers or credit card auto-pay, demonstrate stronger retention. This may indicate that automated payment users are generally more engaged and committed customers.

Key Insights by Visualization

1. Churn Distribution Analysis

- The dataset contains approximately **73.5% retained customers** and **26.5% churned customers**.
 - While the majority of customers remain with the company, more than **1 in every 4 customers leaves**, indicating a substantial retention challenge.
 - The imbalance also suggests that churn prediction models should account for class imbalance during model building.
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2. Gender vs Churn

- The churn percentage among male and female customers is nearly identical.
 - Gender does not appear to significantly influence customer retention behavior.
 - This suggests that customer acquisition and retention strategies do not need gender-specific targeting.
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3. Senior Citizen vs Churn

- Senior Citizens exhibit a noticeably higher churn rate than non-senior customers.
 - Although they represent a smaller portion of the customer base, they contribute disproportionately to churn.
 - This segment may require enhanced customer support, simpler service plans, and proactive engagement initiatives.
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4. Tenure Distribution Analysis

- Customers with shorter tenure periods show the highest churn rates.
- Churn decreases significantly as customer tenure increases.
- Customers who remain beyond the initial months are considerably more likely to become long-term customers.
- The first year of the customer lifecycle represents the highest-risk period for attrition.

Business Impact: Improving onboarding and early customer experience could significantly reduce churn.

5. Contract Type Analysis

- Month-to-Month customers account for the majority of churned customers.
- One-Year and Two-Year contract customers demonstrate substantially higher retention.
- Long-term contracts appear to create stronger customer commitment and lower switching behavior.

Business Impact: Incentivizing annual contracts could directly reduce churn.

13. Payment Method Analysis

- Customers using Electronic Check display the highest churn rates among all payment methods.
- Customers using automatic bank transfers and automatic credit card payments show stronger retention.

- Automated payment methods may indicate higher customer engagement and commitment.

Business Impact: Encouraging customers to switch to automatic payment options could improve retention.

• Key High-Risk Customer Profile

Based on the combined analysis, the customer most likely to churn typically exhibits the following characteristics:

- Month-to-Month contract
- Fiber Optic internet service
- Short tenure with the company
- No Online Security
- No Tech Support
- No Device Protection
- No Online Backup
- Uses Electronic Check as the payment method
- Senior Citizen

• Business Recommendations

To reduce churn and improve customer lifetime value, the company should prioritize:

1. Improving retention during the first 12 months of the customer journey.
2. Encouraging migration from Month-to-Month plans to annual or multi-year contracts.
3. Investigating the root causes of high churn among Fiber Optic customers.
4. Increasing adoption of support and security services through bundling and promotional offers.
5. Promoting automated payment methods to improve customer stickiness.
6. Designing targeted retention programs for Senior Citizens and other high-risk customer segments.

• Conclusion

The analysis demonstrates that churn is driven primarily by **contract type, tenure, internet service category, payment method, and the adoption of support/security services** rather than demographic factors such as gender. Customers receiving greater service value and maintaining longer contractual relationships are considerably more likely to remain with the company. By focusing on these identified risk factors, the organization can significantly improve customer retention and long-term profitability.